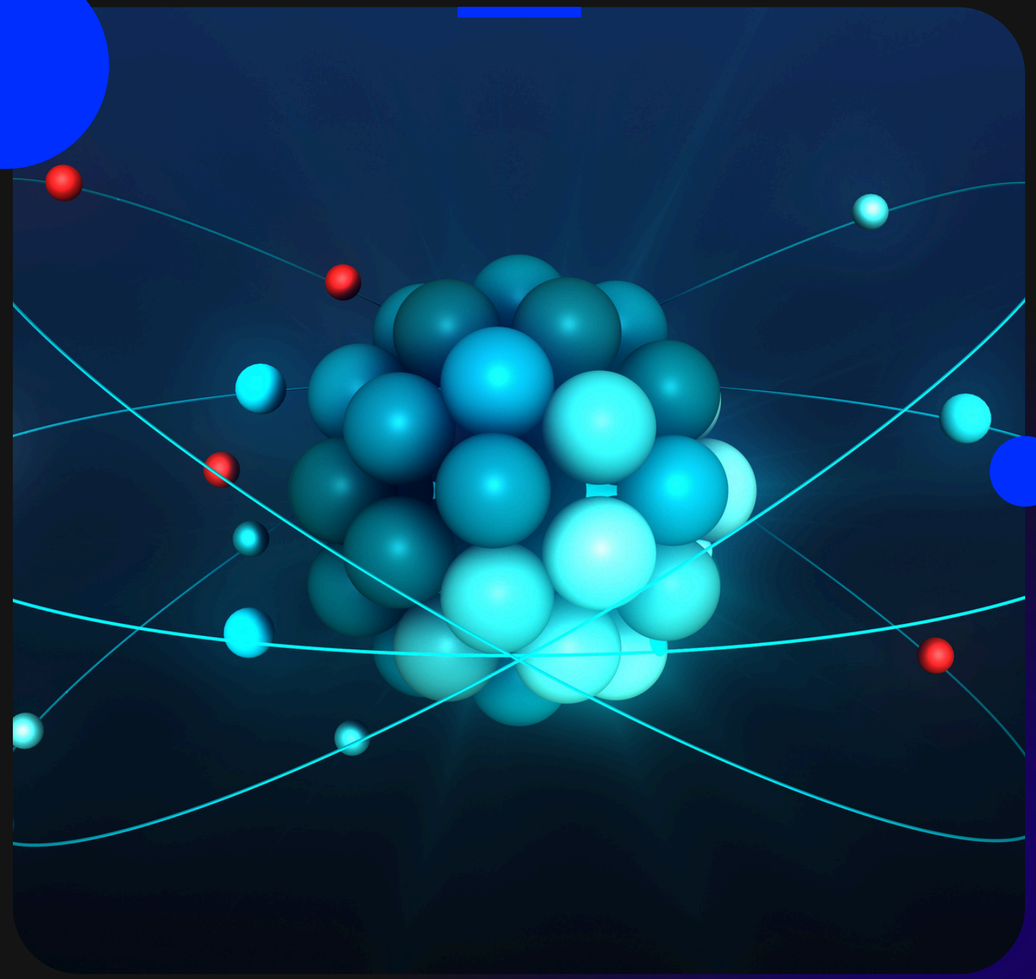
Quantum Key Distribution

Introduction to Quantum Key Distribution



A New Frontier in Technology

- The classical cybersecurity systems are going to be in danger and will be futile when quantum computers advance as time progresses
- This calls for another kind of cryptography, and the quantum way of doing this is “Quantum Key Distribution,” while its classical alternatives are “Post Quantum crypto”.

BB84

The steps involved in developing such a key distribution are:-

1)Quantum communication

2)Phase and sifting.

3)QBER estimation

4)Error reconcilation

5)Privacy Amplification

1)Quantum communication

- Alice sends a sequence of quantum states to Bob .
- Bob measures the qubit under a randomly chosen basis.
- In BB84, the two basis are computational(z) and Hadamard(X).
- Intercepting by an Eavesdropper increases the QBER.

2)Phase and Sifting

- After transmission, Alice and Bob publicly announce the bases used for each qubit.They discard times where bases differ.
- Sifted key: the bit strings where their bases matched.
- Sifting fraction: ideally 1/2 ;can be improved (e.g., asymmetric basis choices).

3)QBER estimation (quantum bit error rate)

- QBER is the fraction of mismatched bits between Alice and Bob..
- Randomly choose a subset of positions from the sifted key, reveal bit values, and compute QBER from this part.
- QBER measures the combined effects of
  Channel noise,
  Detector errors
  Eavesdropping
- QBER can be used to determine how much information Eve could have and thus how much privacy amplification is needed.

4)Error reconcilation

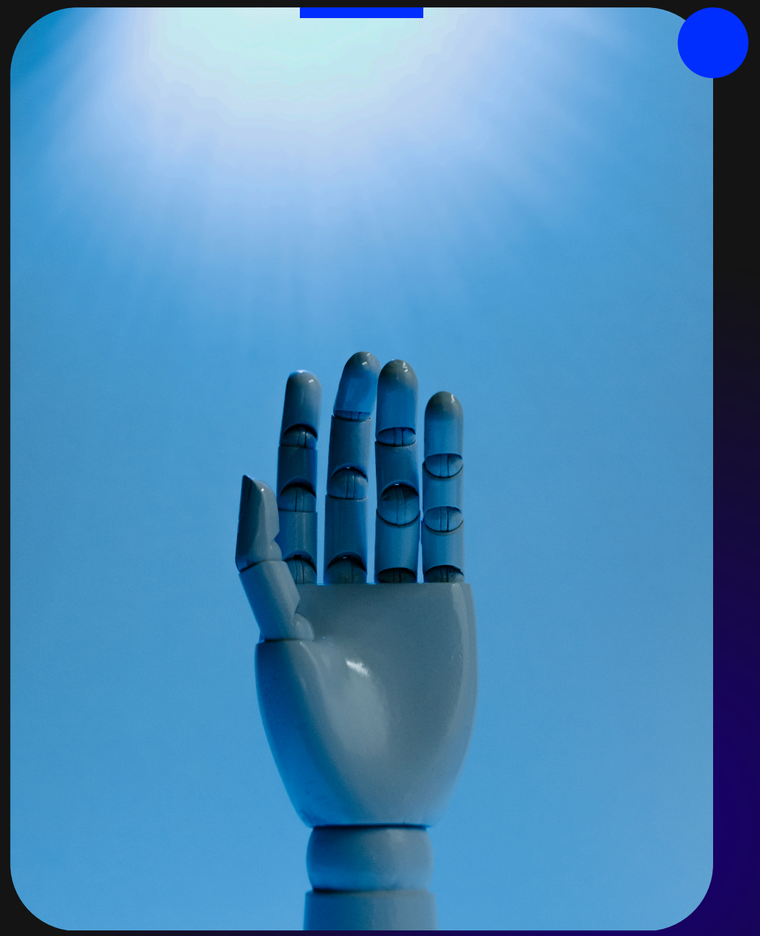
- Turn Alice and Bob’s slightly different sifted strings into identical strings while leaking as little information as possible to Eve.
- Error correction leaks some information; this leak must be accounted for in privacy amplification.
- Performance metric: Actual number of bits revealed compared to the Shannon limit. Shannon limit for a binary symmetric channel with bit error e is

$$H(e) = -e \log_2 e - (1 - e) \log_2 (1 - e)$$

- Practical codes reveal fnH(e) bits with f>=1
- We use the value of f as the performance metric for error reconciliation.
- Typically, f is 1.05–1.2 for good LDPC implementations

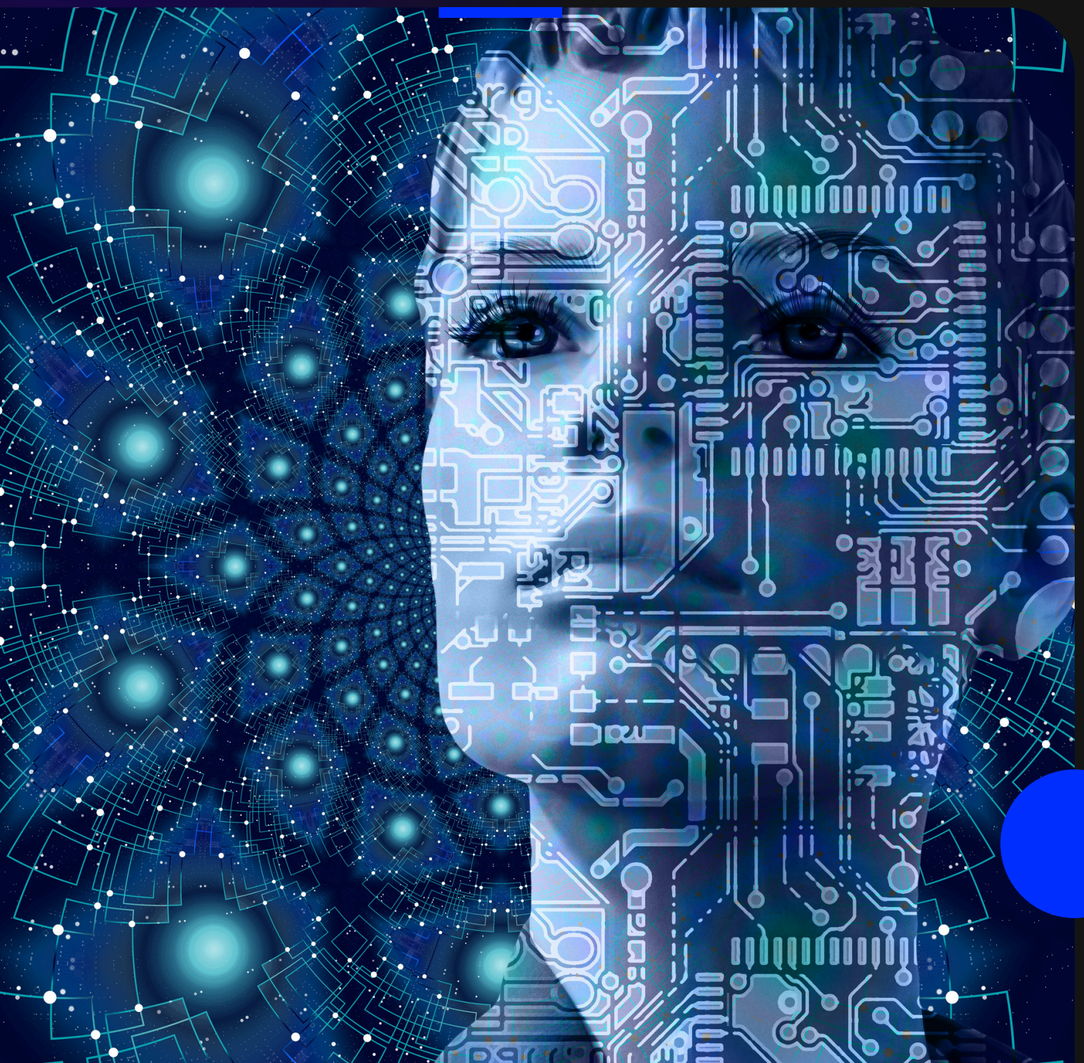
Algorithms for reconcilation

- Cascade: Suitable for small systems, but requires multiple rounds of interaction and leakage that are somewhat larger.
- LDPC-based (one-way): Low interaction, efficient if the code is tuned to the observed e.
- Polar codes.
- Hybrid.



Failure probability

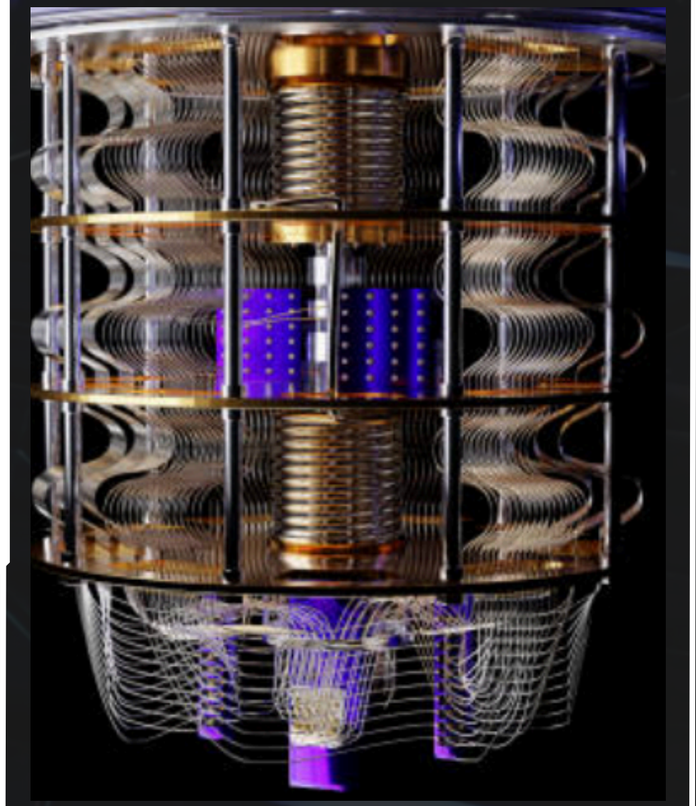
- Reconciliation can fail if Bob is not able to decode. Use CRC/verification. The failure probability must be included as part of the overall security parameter.



5)Privacy Amplification

- Eve may have gained partial information during quantum transmission and definitely learned the reconciliation leakage. Privacy amplification removes her information.
- We use the “Leftover hash lemma ”for this purpose.
- Privacy amplification compresses this reconciled string into a shorter one such that Eve’s knowledge about it becomes negligibly small and the key becomes virtually uniform and secret.

E91



INTRODUCTION TO THE E91 PROTOCOL

- It is an entanglement-based Quantum Key Distribution (QKD) scheme.
- It uses the fundamental principle of Quantum Entanglement to enable the two distant users to generate a shared secret key that is secure against any eavesdropper, even one with a quantum computer.
- The security of E91 is based on the violation of Bell’s Inequality, which demonstrates the presence of genuine quantum entanglement and enables the users to detect any eavesdropper.
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STEPS INVOLVED IN WORKING OF E91

1) Generation of Entangled Photon Pairs
2)Distribution of Entangled Pairs
3)Measurement Process
4)Classical Communication and Sifting
5)Bell's Inequality Test
6)Key Generation

1) GENERATION OF ENTANGLED PHOTON PAIRS

- A source generates pairs of photons that are quantum mechanically entangled in polarization.
- The generation of these pairs can be outsourced to a company
- The presence of an eavesdropper in the company can also be verified using Bell's inequality.
- Each entangled pair exists in a combined quantum state that cannot be described independently; measuring one photon instantly affects the state of the other, no matter the distance.

2)DISTRIBUTION OF ENTANGLED PAIRS

- The source sends one photon from each entangled pair to Alice and the other photon to Bob through quantum channels
- Security in E91 does not depend on trusting the source, which is a major advantage over state-preparation protocols, such as BB84.
- Even if the source is located midway between Alice and Bob or is under the control of an eavesdropper, Eve can't gain useful information. This is because any attempt by Eve to measure or alter the entangled state would break the entanglement and be immediately detectable during the Bell test.

3)MEASUREMENT PROCESS

- ALICE AND BOB EACH RECEIVE ONE PHOTON FROM THE ENTANGLED PAIR AND MEASURE IT USING A POLARIZATION ANALYZER.
- THEY EACH RANDOMLY SELECT FROM A PREDEFINED SET OF MEASUREMENT BASES (FOR INSTANCE, THREE ANGLES FOR ALICE AND FOR BOB).
- AS THE PHOTONS ARE ENTANGLED, ALICE’S AND BOB’S MEASUREMENT OUTCOMES WILL SHOW CORRELATED PATTERNS, EVEN THOUGH THEIR CHOICES OF MEASUREMENT SETTINGS ARE INDEPENDENT.
- THE RANDOM CHOICE OF MEASUREMENT BASES ENSURES THAT ANY EAVESDROPPER CANNOT PREDICT THE SETTINGS AND THEREFORE CANNOT EXTRACT INFORMATION WITHOUT DISTURBING THE CORRELATIONS.

4)CLASSICAL COMMUNICATION AND SIFTING

- AFTER COMPLETING THE QUANTUM TRANSMISSION AND MEASUREMENTS, ALICE AND BOB COMMUNICATE OVER A PUBLIC CLASSICAL CHANNEL, WHICH CAN BE OBSERVED BUT NOT ALTERED.
- THEY ANNOUNCE THE MEASUREMENT BASES USED FOR EACH PHOTON BUT DO NOT REVEAL THEIR ACTUAL MEASUREMENT RESULTS.
- THEY THEN DISCARD THE DATA WHERE THEIR BASES WERE INCOMPATIBLE AND RETAIN THE OUTCOMES CORRESPONDING TO COMPATIBLE SETTINGS.
- THE CORRELATED BITS WE GET FROM THIS DATA FORM THE RAW KEY, WHICH WILL LATER BE PROCESSED TO PRODUCE THE FINAL SECURE KEY.
- EVEN IF EVE LISTENED TO THE CLASSICAL COMMUNICATION, SHE GAINS NO INFORMATION ABOUT THE ACTUAL KEY VALUES.

5)Bell’s Inequality Test

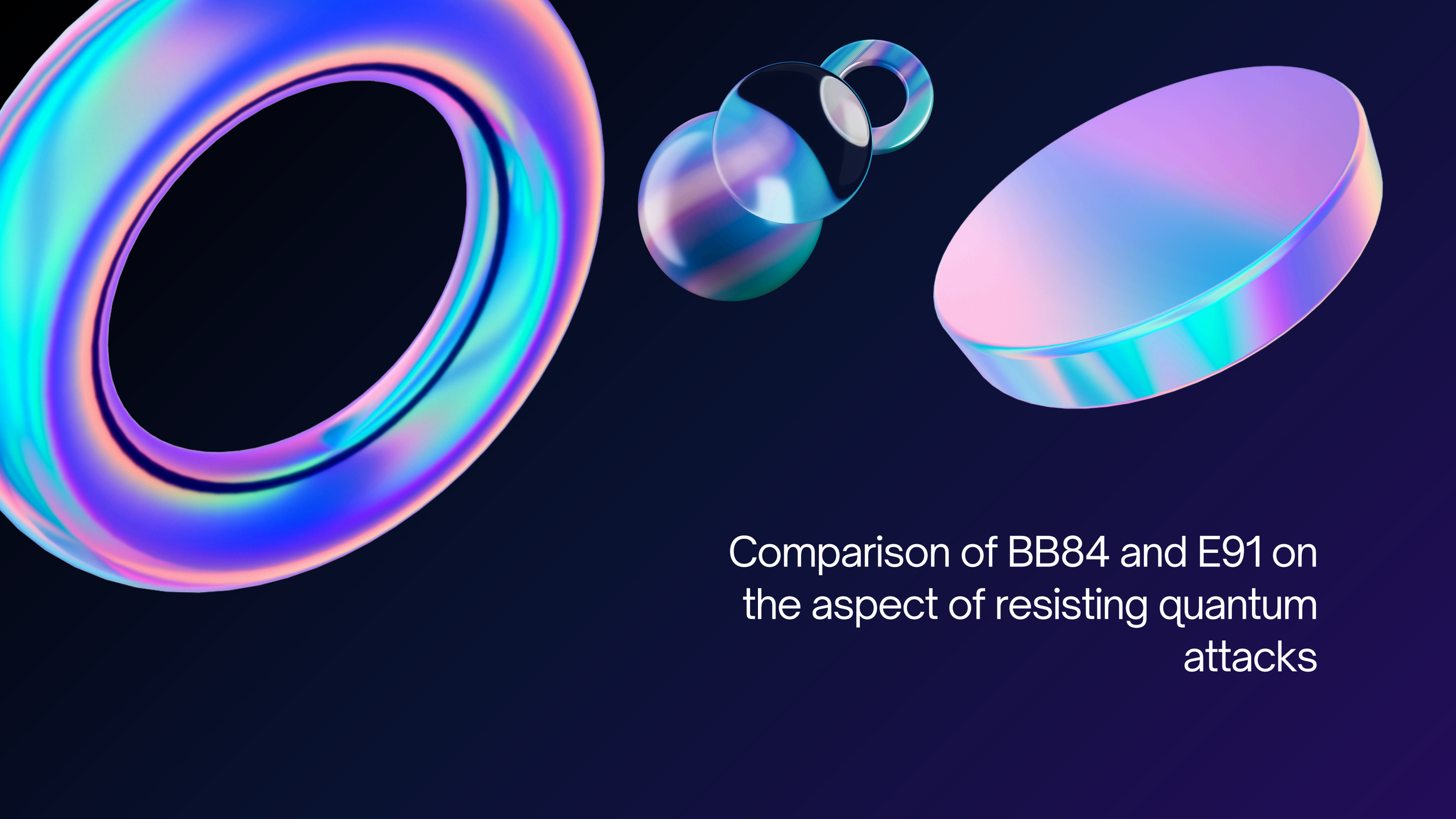
- A CRUCIAL STEP IN THIS PROTOCOL WOULD BE TO VERIFY THE PRESENCE OF TRUE QUANTUM ENTANGLEMENT THROUGH BELL’S INEQUALITY TESTING.
- ALICE AND BOB USE A SELECTED SUBSET OF THEIR RESULTS TO CALCULATE THE BELL PARAMETER (S).
- IF THE VALUE OF S EXCEEDS 2, IT INDICATES A VIOLATION OF BELL’S INEQUALITY,
- THIS RESULT PROVES THAT NO EAVESDROPPER COULD HAVE CREATED OR SHARED THESE CORRELATIONS WITH ALICE AND BOB.
- IF THE BELL TEST FAILS (S ≤ 2), THEY ABORT THE PROTOCOL, AS IT MEANS THE ENTANGLEMENT WAS LOST OR TAMPERED WITH.

6)KEY GENERATION

ONCE THE SECURITY OF THE QUANTUM CHANNEL IS VERIFIED THROUGH THE BELL TEST, WE CAN USE THE REMAINING BITS TO GENERATE THE SECRET KEY.

THE KEY GENERATION PROCESS INVOLVES TWO IMPORTANT STAGES:

1. ERROR CORRECTION: BOTH PARTIES EMPLOY CLASSICAL ALGORITHMS TO CORRECT MINOR DISCREPANCIES CAUSED BY NOISE OR MEASUREMENT IMPERFECTIONS.
2. PRIVACY AMPLIFICATION: A MATHEMATICAL PROCEDURE THAT COMPRESSES THE KEY IN SUCH A WAY THAT EVEN IF AN EAVESDROPPER OBTAINED PARTIAL INFORMATION, HER KNOWLEDGE BECOMES NEGLIGIBLE.



Comparison of BB84 and E91 on the aspect of resisting quantum attacks

- BB84: Uses single photons prepared in one of four polarization states i.e., |0>,|1>,|+>,|->.
- E91: Uses entangled photon pairs shared between Alice and Bob this is also a feature where Alice can outsource the task of random number generating for keys to a company, as well as check for the presence of an eavesdropper in the company by the following:-
- $$S = E(a_1, b_1) - E(a_1, b_3) + E(a_3, b_1) + E(a_3, b_3)$$

  In the ideal case, S>2 , typically around 2.828
- In case of eavesdropper the value drops below 2 .

BB84 against quantum attacks

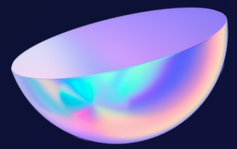
- Based on the no-cloning theorem and measurement disturbance.
- Even if Eve has a quantum computer:

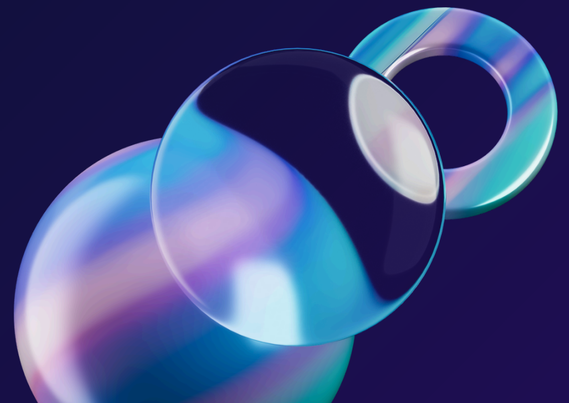
  She cannot perfectly distinguish non-orthogonal states (e.g., |0⟩ vs |+⟩).
- Her presence can always be detected, taking QBER as a parameter.
- Privacy amplification removes the possibility of Quantum Memory Attacks
- Thus, BB84 resists quantum attacks by relying on:

Quantum uncertainty
Error checking
Privacy amplification

E91 against quantum attacks

- Based on entanglement and Bell's inequality violation.
- Entanglement-based attacks are detected by using Bell's inequality as discussed earlier.
- Better than BB84 in the aspect of source tampering.
- E91 can be extended to Device-Independent Quantum Key Distribution, making it immune to Measurement-device attacks.
- Immune to Quantum Memory Attacks due to nonlocality.

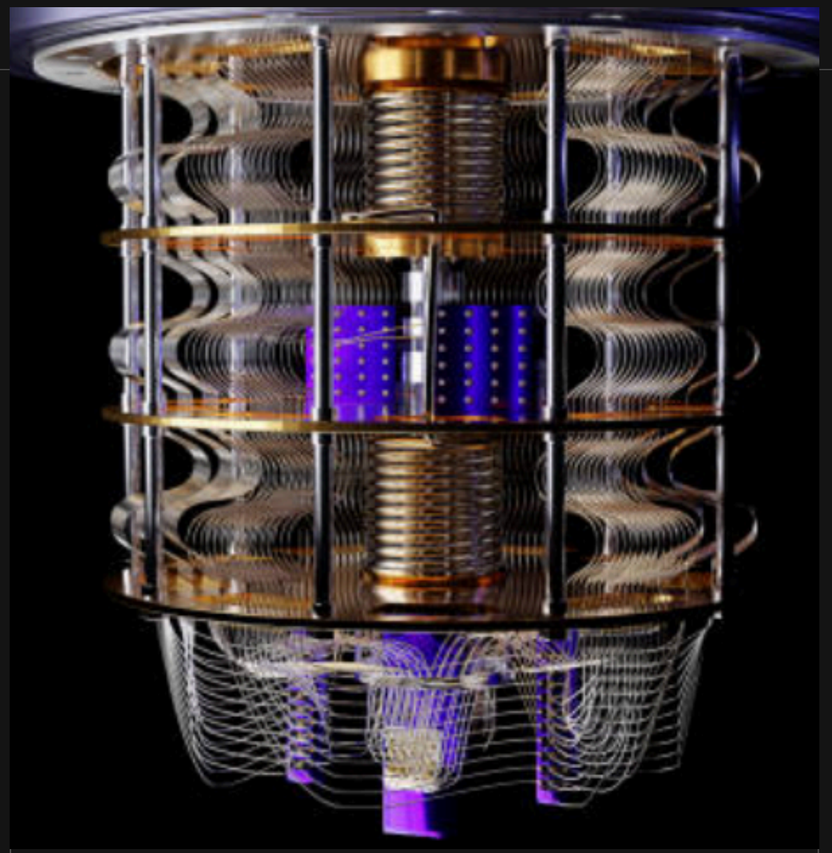




Overview of BB84 vs E91 in terms of resistance against Quantum Attacks

- BB84 is simple in terms of operation and secure against quantum computational attacks, but it assumes trusted devices and ideal single photons.
- E91, on the other hand, is conceptually stronger as it proves security from the observed quantum correlations themselves, removing some trust assumptions.
- In resisting quantum attacks, E91 > BB84, because E91 can remain secure even if Eve has control over the source or measurement devices, as long as Bell violations are observed.

Thank You



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